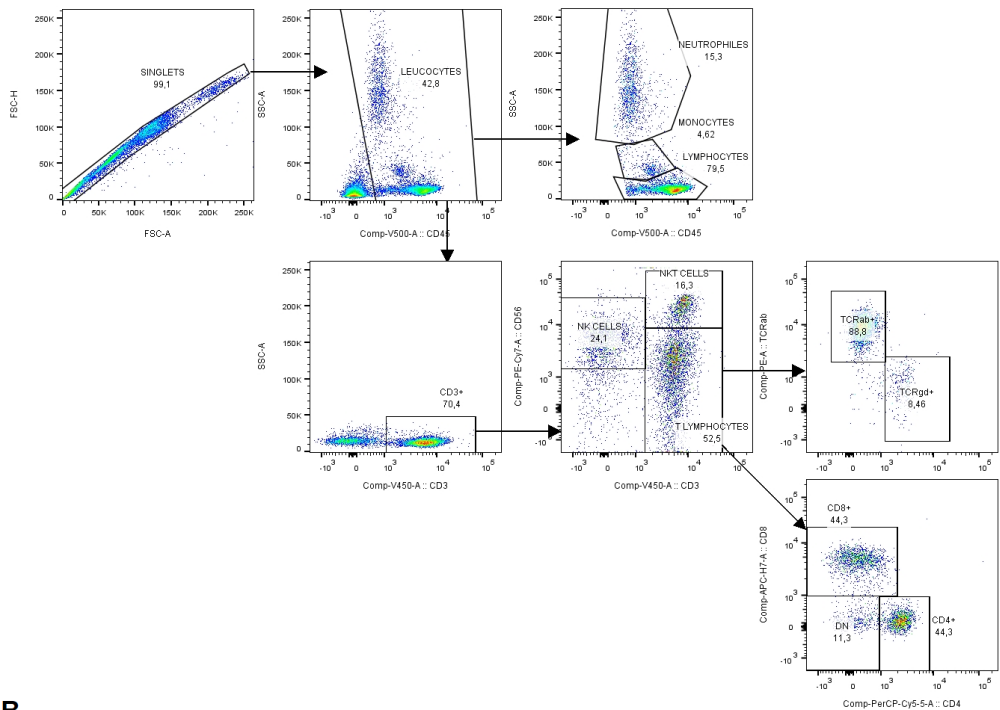


Supplementary Material

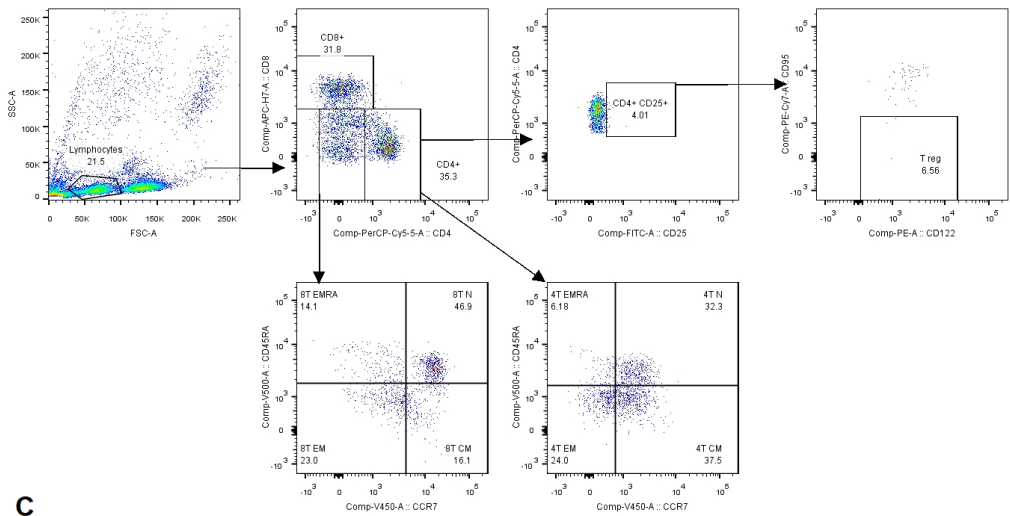
1 Supplementary Figures and Tables

1.1 Supplementary Figures

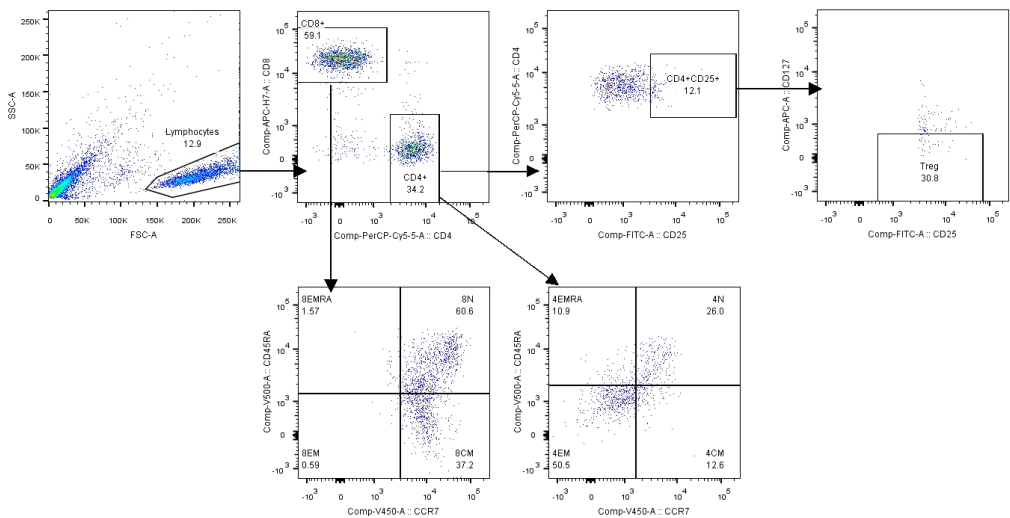
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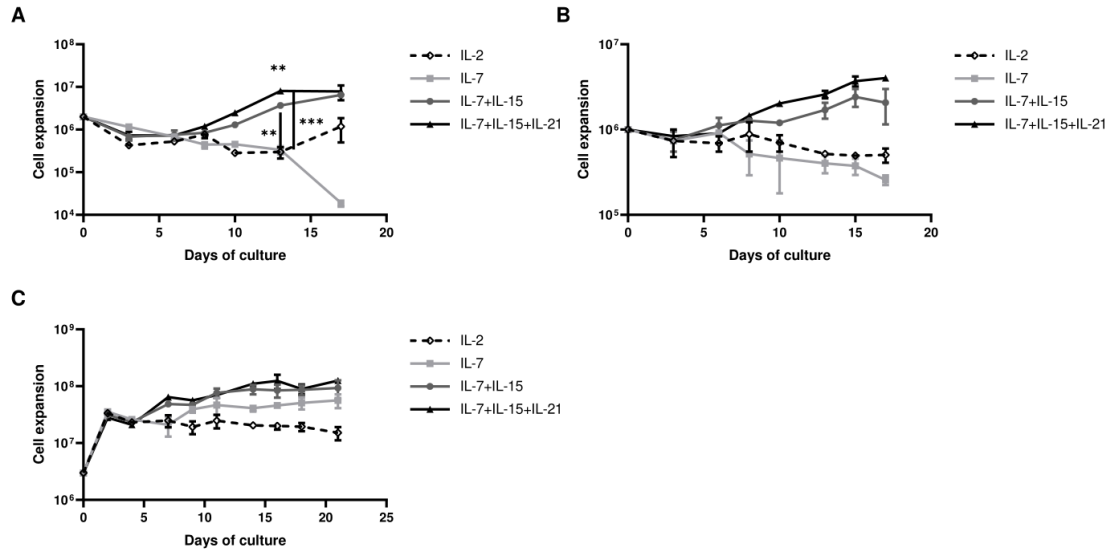
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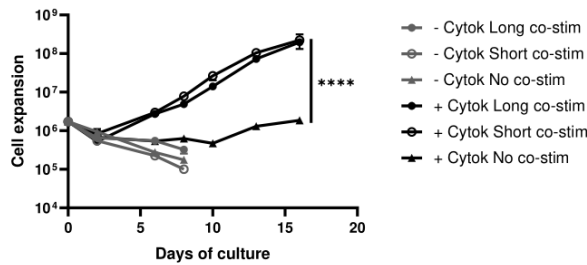
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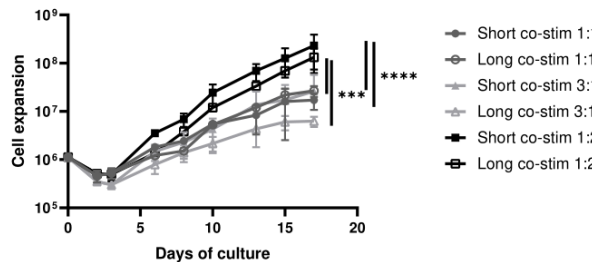
Supplementary Figure 1. Gating strategy for identification of T cell subsets. **(A)** Gating strategy for identification of T lymphocytes from different samples (PB, 45RA+ and CB) at before culture. **(B)** Gating strategy for the identification of different T lymphocyte subsets (T_N , T_{CM} , T_{EM} and T_{EMRA}) in PB and CB samples before culture. **(C)** Gating strategy for identification of different T cell subpopulation after 14-day culture. CB, cord blood; PB, peripheral blood; 45RA, CD45RA+ fraction post-apheresis. Treg, Regulatory T lymphocytes; T_N , naïve T cell; T_{CM} , central memory T cells; T_{EM} , effector memory T cells; T_{EMRA} , terminal effector T cells.



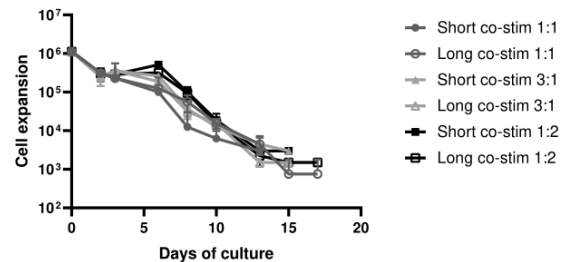
Supplementary Figure 2. Proliferation of T cells in culture under different cytokine conditions (IL-2, IL-7, IL-7+IL-15 or IL-7+IL-15+IL-21). **(A)** Proliferation of PB-derived lymphocytes during 17 days of culture. **(B)** Proliferation of 45RA+ post-apheresis-derived lymphocytes during 17 days of culture. **(C)** Proliferation of CB-derived lymphocytes during 21 days of culture. **p value < 0.005. ***p value < 0.0005. CB, cord blood; PB, peripheral blood.

A**B**

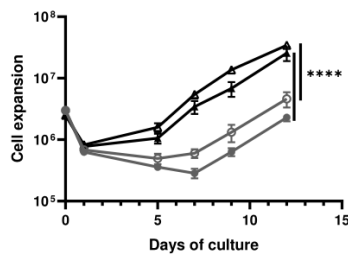
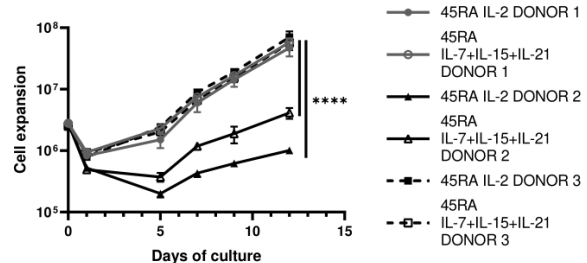
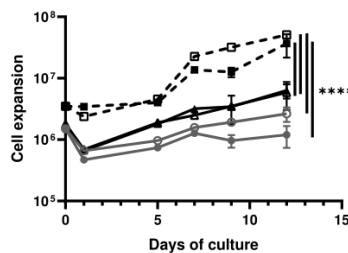
Lymphocyte proliferation with cytokines (IL-7+IL-15)

**C**

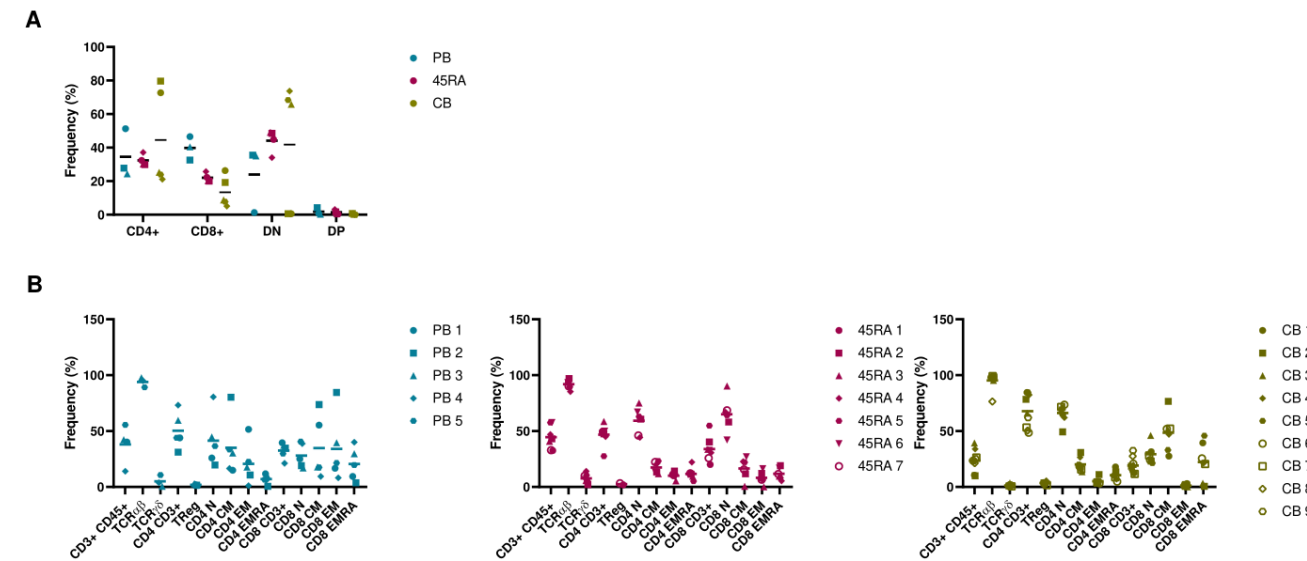
Lymphocyte proliferation without cytokines



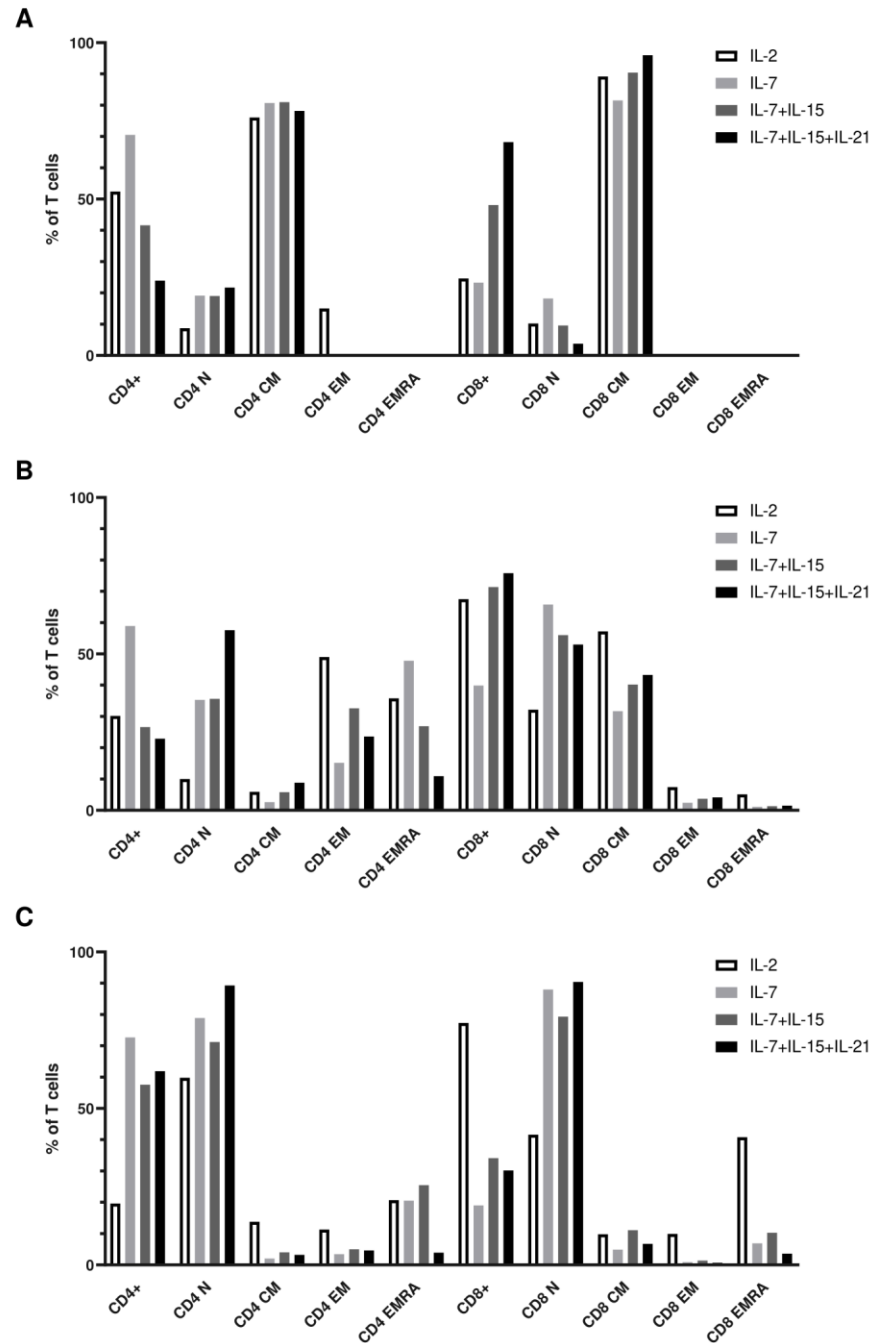
Supplementary Figure 3. Proliferation of peripheral blood T cells with or without beads and cytokines. **(A)** Lymphocyte proliferation in different culture conditions with or without cytokines and anti-CD3/CD28 co-stimulus. **(B)** Lymphocyte proliferation without cytokines and short (48 hours) or long (all culture period) co-stimulation with beads and different bead:T cell ratios (1:1, 3:1, 1:2). **(C)** Lymphocyte proliferation in culture with IL-7+IL-15 cytokines and short/long co-stimulation with beads at different bead:T cell ratios (1:1, 3:1, 1:2). ***p value < 0.0005. ****p value < 0.0001.

A**B****C**

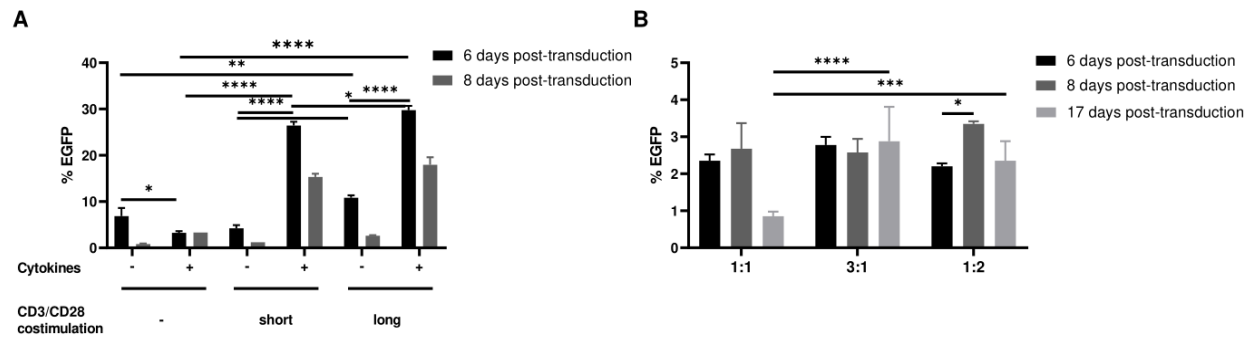
Supplementary Figure 4. Proliferation of lymphocytes from different donors cultured with IL-2 or IL-7+IL-15+IL-21. **(A)** Proliferation of PB-derived lymphocytes from 2 different healthy donors. **(B)**



Supplementary Figure 5. Phenotype of T cells from PB, 45RA and CB before culture. **(A)** Cell phenotype according to CD4 and CD8 markers. DN: double negative (CD4-CD8-). DP: double positive (CD4+CD8+). **(B)** Different T cell subsets assessed by flow cytometry. CB, cord blood; PB, peripheral blood; 45RA, CD45RA+ fraction post-apheresis.

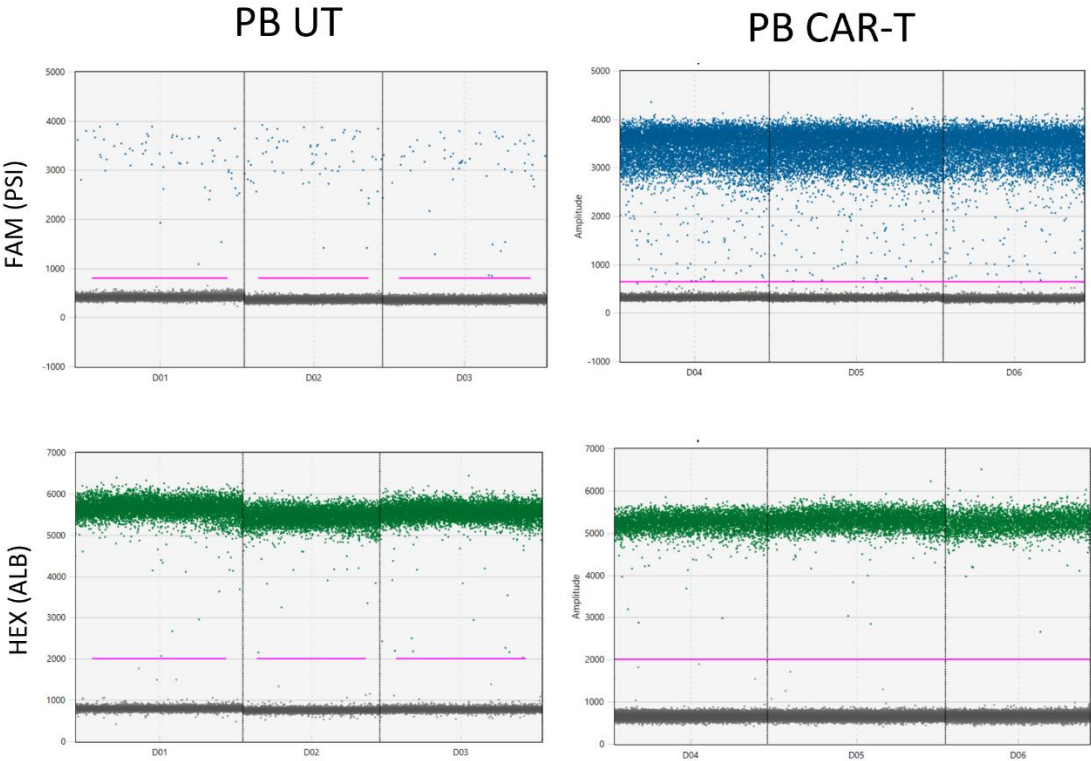


Supplementary Figure 6. Phenotype of T cell subsets in CD4+ and CD8+ lymphocytes after 10 days in culture under different cytokine conditions (IL-2, IL-7, IL-7+IL-15 or IL-7+IL-15+IL-21). **(A)** Phenotype of cells derived from PB. **(B)** Phenotype of T cells derived from 45RA fraction after apheresis. **(C)** Phenotype of T cells derived from CB. CB, cord blood; PB, peripheral blood; 45RA, CD45RA+ fraction post-apheresis.

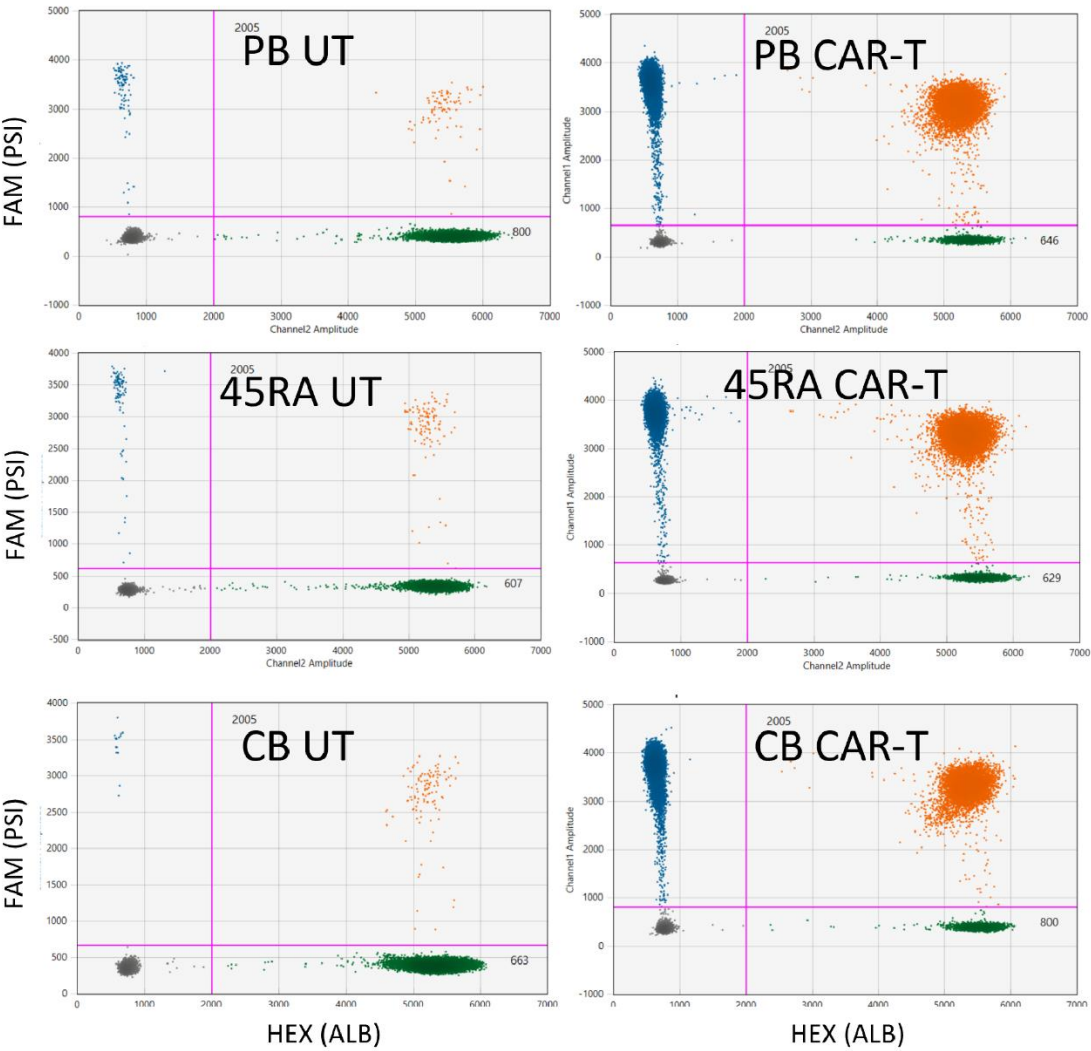


Supplementary Figure 7. Transduction efficiency of an EGFP-expressing LV in T cells in different culture conditions. **(A)** Lymphocyte transduction in culture with or without cytokines and beads co-stimulation. **(B)** Lymphocyte transduction in culture with different ratios (1:1, 3:1, 1:2) of beads co-stimulus. *p value < 0.05. **p value < 0.005. ***p value < 0.0005. ****p value < 0.0001.

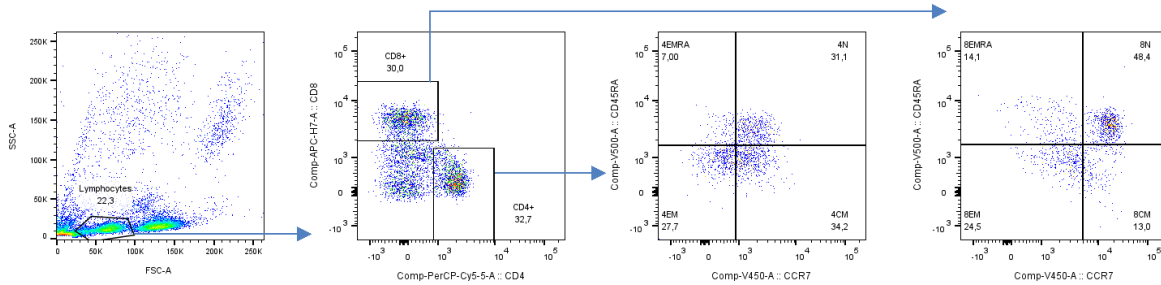
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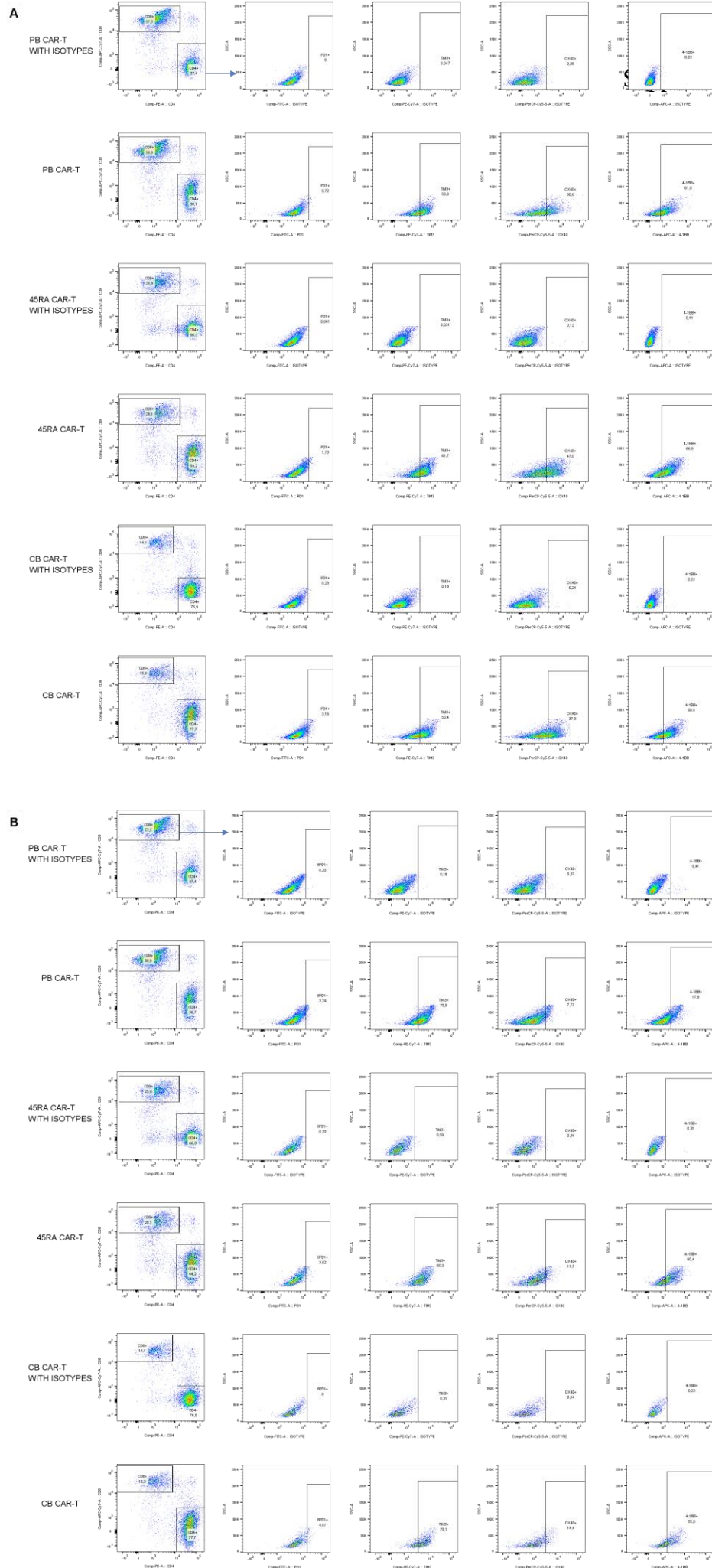
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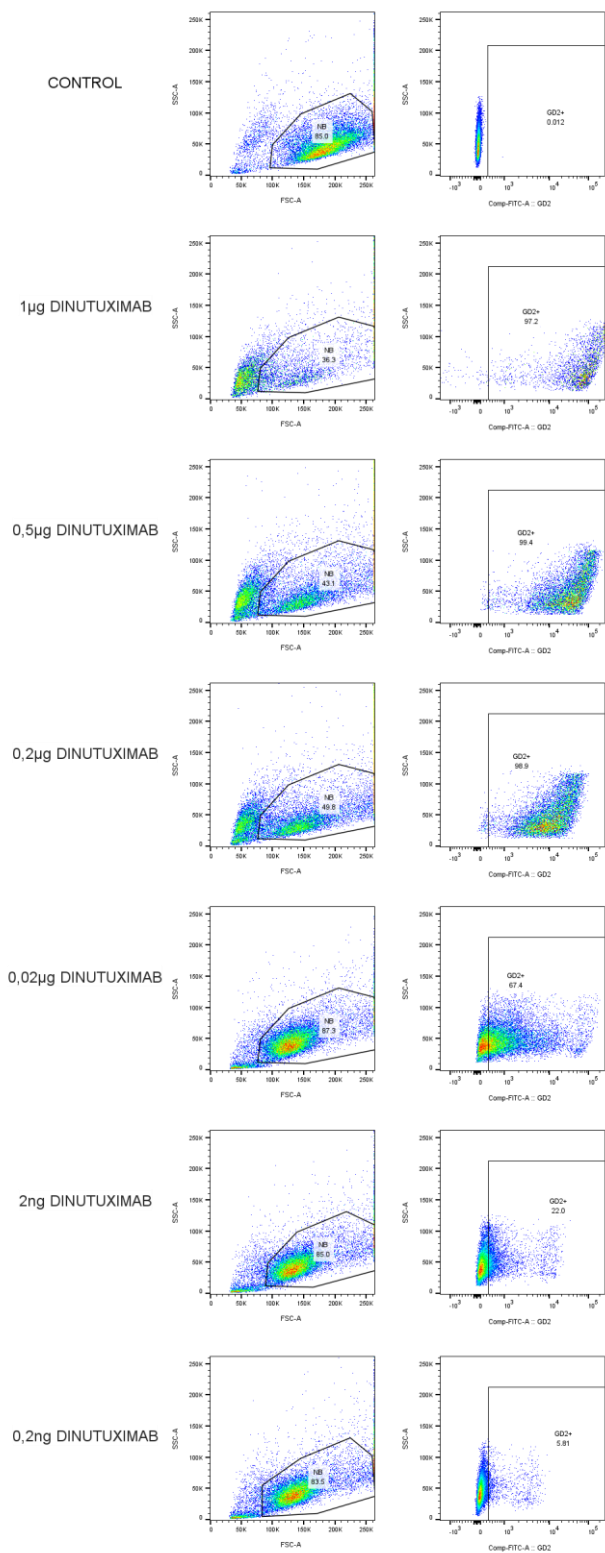
Supplementary Figure 8. Vector copy number by ddPCR. **(A)** Spikes of Psi signal in FAM channel and albumin gene in HEX channel of untransduced PB T cells (PB UT) and PB CAR-T samples. **(B)** Representative data for thresholds set at 600 for psi signal (FAM channel in blue) and 2000 for albumin internal control (HEX channel in green) in 2D amplitude of CAR-T derived from PB, 45RA and CB, and the controls of untransduced T cells. CB, cord blood; PB, peripheral blood; 45RA, CD45RA+ fraction post-apheresis.



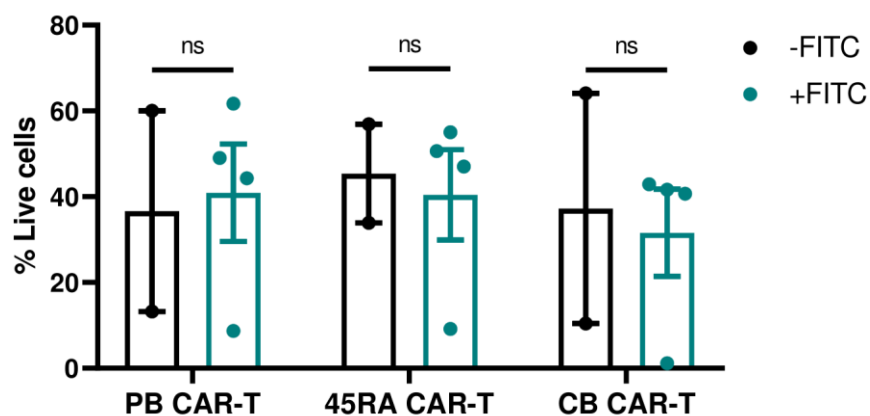
Supplementary Figure 9. PBMCs before treatment stained with antibodies for the identification of T lymphocyte subsets: naïve T cells (T_N), central memory T cells (T_{CM}), effector memory T cells (T_{EM}) and terminal effector T cells (T_{EMRA}).



Supplementary Figure 10. Representative dot plots of exhaustion status of CAR-T cells prior co-cultures including controls with isotype stanning. **(A)** Dot plots of CD4+ CAR-T cells. **(B)** Dot plots of CD8+ CAR-T cells.



Supplementary Figure 11. Representative dot plots of LAN-1 cell line stained with different concentrations of Dinutuximab conjugated to FITC (1 μ g, 0.5 μ g, 0.2 μ g, 0.02 μ g, 2ng and 0.2ng).



Supplementary Figure 12. LAN-1 cell viability (non-normalized) assessed by flow cytometry in co-culture with CAR-T cells at 2:1 CAR:NB ratio.

1.2 Supplementary Tables

Supplementary Table 1. MQR files of each sample processed for OGM (separate pdf document).

Supplementary Table 2. Structural variants (SVs) found in each sample (control samples at day 0 and CAR-T products at day 7) (separate excel document).